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Department of CSE

B.Tech. CSE Trimester -5

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Generative AI
25CS2207E

LAB RECORD

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Aim:

To implement and compare three subword tokenization approaches—Byte Pair Encoding (BPE), WordPiece and SentencePiece—using English, Telugu and Hindi text, and observe how each tokenizer splits text into tokens and how many tokens are produced.

Software requirements:

1. Python 3.10 or later.
2. Visual Studio Code (VS Code).
3. An active Python virtual environment.
4. Internet connection to download pretrained tokenizer files.
5. Required Python libraries:
    - Transformers
    - SentencePiece

Python Program:

tokenizer_comparison.py

```
from transformers import AutoTokenizer


# Define the three tokenizers
TOKENIZERS = {
    "BPE - GPT-2": "gpt2",
    "WordPiece - Multilingual BERT": "bert-base-multilingual-cased",
    "SentencePiece - XLM-R": "xlm-roberta-base"
}
```

```
# Input sentences in three languages
TEXTS = {
    "English": "Artificial intelligence is changing the world.",
    "Telugu": "కృత్రిమ మేధస్సు ప్రపంచాన్ని మారుస్తోంది.",
    "Hindi": "कृत्रिम बुद्धिमत्ता दुनिया को बदल रही है।"
}


# Load pretrained tokenizers
def load_tokenizers():
    loaded = {}

    for display_name, model_name in TOKENIZERS.items():
        print(f"Loading {display_name}...")
        loaded[display_name] = AutoTokenizer.from_pretrained(model_name)

    return loaded


# Compare the tokenizers
def compare_tokenizers(loaded_tokenizers):
```

```
for language, text in TEXTS.items():

    print("\n" + "=" * 75)
    print(f"LANGUAGE: {language}")
    print(f"INPUT TEXT: {text}")
    print("=" * 75)

    for tokenizer_name, tokenizer in loaded_tokenizers.items():

        encoded = tokenizer(
            text,
            add_special_tokens=True,
            truncation=False
        )

        token_ids = encoded["input_ids"]
        tokens = tokenizer.convert_ids_to_tokens(token_ids)

        print(f"\n{tokenizer_name}")
        print("Tokens     :", tokens)
        print("Token IDs  :", token_ids)
```

```
        print("Token count:", len(token_ids))


# Main function
def main():
    tokenizers = load_tokenizers()
    compare_tokenizers(tokenizers)


if __name__ == "__main__":
    main()
```

Output:

```
================================================================================
LANGUAGE: English
INPUT TEXT: Artificial intelligence is changing the world.
================================================================================

BPE - GPT-2
Tokens     : ['Art', 'ificial', 'Ġintelligence', 'Ġis', 'Ġchanging', 'Ġthe', 'Ġworld', '.']
Token IDs  : [8001, 9542, 4430, 318, 5609, 262, 995, 13]
Token count: 8

WordPiece - Multilingual BERT
Tokens     : ['[CLS]', 'Arti', '##ficial', 'intelligence', 'is', 'changing', 'the', 'world', '.', '[SEP]']
Token IDs  : [101, 90841, 69229, 30151, 10124, 43068, 10105, 11356, 119, 102]
Token count: 10

SentencePiece - XLM-R
Tokens     : ['<s>', '_Artificial', '_', 'intelligence', '_is', '_changing', '_the', '_world', '.', '</s>']
Token IDs  : [0, 231996, 6, 130687, 83, 151134, 70, 8999, 5, 2]
Token count: 10
```

```
================================================================================
LANGUAGE: Telugu
INPUT TEXT: కృత్రిమ మేధస్సు ప్రపంచాన్ని మారుస్తోంది.
================================================================================

BPE - GPT-2
Tokens     : ['à°', 'ĸ', 'à±', 'Ĥ', 'à°', '¤', 'à±', 'ī', 'à°', '°', 'à°', '¿', 'à°', '®'
Token IDs  : [156, 108, 243, 156, 109, 225, 156, 108, 97, 156, 109, 235, 156, 108, 108, 156, 108, 123, 156, 108, 106,
Token count: 112

WordPiece - Multilingual BERT
Tokens     : ['[CLS]', 'క', '##ృ', '##త్రి', '##మ', 'మ', '##ే', '##ధ', '##స్సు', '##ను', 'ప్రపంచ', '##ాన్ని', 'మ', '##ార
Token IDs  : [101, 1201, 111339, 88046, 24696, 1224, 17729, 99288, 15370, 39179, 94504, 23769, 1224, 27374, 15370, 68
Token count: 18

SentencePiece - XLM-R
Tokens     : ['<s>', '▁', 'కృత', 'ి', 'మ', '▁మే', 'ధ', 'స్సు', '▁ప్రపంచ', 'ాన్ని', '▁మారు', 'స్తోంది', '.', '</s>']
Token IDs  : [0, 6, 151424, 91445, 3071, 22290, 11600, 129461, 86457, 6258, 169610, 104528, 5, 2]
Token count: 14

================================================================================
```

```
================================================================================
LANGUAGE: Hindi
INPUT TEXT: कृत्रिम बुद्धिमत्ता दुनिया को बदल रही है।
================================================================================

BPE - GPT-2
Tokens     : ['à¤', 'ĸ', 'à¥', 'Ĥ', 'à¤', '¤', 'à¥', 'ī', 'à¤', '°', 'à¤', '¿', 'à¤', '®', 'Ġà¤', '¬', '
Token IDs  : [11976, 243, 24231, 225, 11976, 97, 24231, 235, 11976, 108, 11976, 123, 11976, 106, 28225,
Token count: 68

WordPiece - Multilingual BERT
Tokens     : ['[CLS]', 'क', '##ृत', '##्', '##िम', 'ब', '##ुद्ध', '##ि', '##त्ता', 'दुनिया', 'को', 'ब', '##दल
Token IDs  : [101, 865, 37500, 18321, 50419, 887, 48010, 50419, 94034, 83874, 11267, 887, 99989, 57203,
Token count: 17

SentencePiece - XLM-R
Tokens     : ['<s>', '▁कृत', 'ि', 'म', '▁बुद्धि', 'म', 'त्ता', '▁दुनिया', '▁को', '▁बदल', '▁रही', '▁है', '।', '</s>'
Token IDs  : [0, 74707, 97885, 1920, 105202, 1920, 54297, 29405, 629, 14512, 8906, 460, 125, 2]
Token count: 14

Process finished with exit code 0
```